HESHAM ELSHIMY

SOFTWARE ENGINEER

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SUMMARY

Backend Engineer focused on Node.js, Golang and core fundamentals, building scalable backend systems and passionate about teaching and simplifying complex concepts. First place in competitive programming at my university.

PROFESSIONAL EXPERIENCE

Software Engineering Intern | Sprints

2026 · 2 months

- Completed a 1-month virtual internship working on PitIQ, contributing to the development of a full-stack quiz platform.
- Recognized as a Top Achiever and selected for an additional 1-month internship working with Sprints' engineering team.
- Contributed to Sprints' production Learning Management System (LMS), serving 250k+ users.
- Fixed bugs and implemented small features in a large, established production codebase.
- Worked with existing architecture and business logic while maintaining compatibility with a live production system.
- Served in a dual developer/mentor capacity — mentored incoming teams of developers, helping them organize their projects and guiding their technical approach.
- Technologies: NestJS, Prisma, PostgreSQL, PHP, Laravel, Blade.

Course Content Creator (Golang) | CS Skilled Academy

Under the supervision of Dr. Mostafa Saad (Senior Computer Vision Engineer, NVIDIA)

- Developed intensive Go (Golang) course covering fundamentals, **concurrency**, advanced topics

Coding Instructor & Educational Head | U-Code Academy

- Managed and supported instructors to ensure high-quality teaching
- Designed and delivered "Code Mystery" courses (Level 1 & 2) with C++

Coding Instructor for Kids | Robus Academy

- Taught coding basics using Scratch
- Delivered interactive lessons on logic and problem-solving

LEADERSHIP AND ACHIEVENENTS

Vice President and Problem Solving Instructor | ICPC ECU SB

- Designed and delivered training sessions for students preparing for the Egyptian Collegiate Programming Contest (ECPC), focusing on algorithmic problem-solving and competitive programming techniques.

1st Place __Competitive Programming, Egyptian Chinese University (university-level)

PROJECTS

PitIQ – Educational Quiz Platform | Next.js, NestJS · Team Project

- Owned the UI identity and design system for an educational quiz/assessment SaaS platform built for academies, training centers, schools, and universities
- Built the full frontend of the Student Portal, including the quiz-taking experience, dashboard, and results views, using Next.js
- Built the backend orchestrator layer coordinating quiz flow and core business logic using NestJS

Code Security Identifier | Python · Team Project

- Built an AI-powered source-code vulnerability detection tool using a **dual-encoder** architecture that jointly processes code through GraphCodeBERT and VulBERT.

SKILLS

Languages: JavaScript, TypeScript, C++, GoLang,PHP

Backend: Node.js, Express, NestJS,Laravel REST APIs, MVC Architecture

Databases: PostgreSQL, MySQL

Fundamentals: Data Structures & Algorithms, OOP, Design Patterns, SOLID Principles

Tools: Git, Postman, Docker

EDUCATION

Egyptian Chinese University — Cairo, Egypt (2022 – now)

- Major: Engineering, Software & Information Technology
- GPA: 3.6